

NECO

Mathematics Syllabus

A Comprehensive Guide To Acing Your NECO Examinations With Ease



NECO
Standard

Includes:

- Syllabus
- Recommended Textbooks
- NECO Grading System

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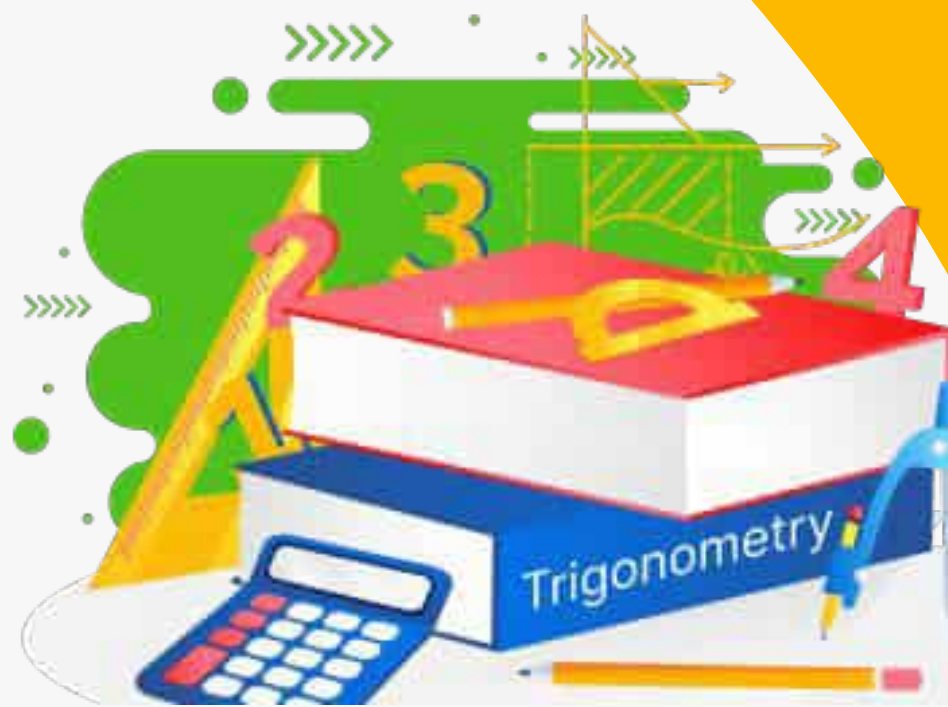


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ABOUT US

Syllabus NG is the premier destination for all educational resources and exam preparation materials in Nigeria and the world. Our mission is simple yet impactful; to empower learners of all ages and backgrounds with the resources they need to succeed academically. It is always better to work smarter than to work harder.

We offer educational consulting, research, and counselling services for individuals, schools, and institutions.

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ABOUT THE NATIONAL EXAMINATION COUNCIL (NECO)

The National Examination Council (NECO) is an examination board that conducts the Senior Secondary Certificate Examination (SSCE) for internal and external candidates within Nigeria. This examination board also conducts other examinations such as the National Common Entrance Examination (NCEE) and Basic Education Certificate Examination (BECE).

The SSCE conducted by NECO is an examination written by students who have concluded their senior secondary education. The certificate is one of the prerequisite documents for gaining admission into a tertiary institution for students who are willing to further their education.

The BECE is an examination written by students who have completed their junior secondary education and are willing to advance to the senior secondary level.

The NCEE is an exam taken by students in their final year of primary education, typically in the 6th year.

NECO has an online result verification/confirmation system where students can check their SSCE, BECE, and NCEE results, using purchased tokens.

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Chapter One Aims & Objectives

About

Mathematics is one of the four important subjects in the NECO exams. Without a credit in it, you can't get admitted into a higher institution.

Having a trusted guide who will give you all the insider tips on how the exam will look is a must. With this syllabus, you get just that.

By studying this syllabus, you're not just prepping for an exam; you are also laying down a solid math foundation for life. Imagine confidently tackling everyday challenges, from algebraic expressions to statistics.

This NECO General Mathematics syllabus is your ticket to success in your math exam so download and study.

Marking Guides & Sections

There will be two papers, Papers 1 and 2, both of which must be taken.

PAPER 1:

This will consist of fifty multiple-choice objective questions, drawn from the common areas of the syllabus, to be answered in 1½ hours for 50 marks.

PAPER 2:

It will consist of thirteen essay questions in two sections – Sections A and B, to be answered in 2½ hours for 100 marks. Candidates will be required to answer ten questions in all.

Section A -Will consist of five compulsory questions, elementary carrying a total of 40 marks. The questions will be drawn from the common areas of the syllabus.

Section B - Will consist of eight questions of greater length and difficulty. The questions shall include a maximum of two which shall be drawn from parts of the syllabuses which may not be peculiar to candidates' home countries.

Candidates will be expected to answer five questions for 60 marks.

Chapter Two

Syllabus

1	NUMBER AND NUMERATION	1.1 Number bases i. conversion of numbers from one base to another ii. Basic operations on number bases 1.2 Modular Arithmetic (i) Concept of Modulo Arithmetic. (ii) Addition, subtraction and multiplication operations in modulo arithmetic. (iii) Application to daily life. 1.3 Fractions, Decimals and Approximations (i) Basic operations on fractions and decimals. (ii) Approximations and significant figures 1.4 Indices (i) Laws of indices (ii) Numbers in standard form (scientific notation) 1.5 Logarithms (i) Relationship between indices and logarithms e.g. $y = 10^k$ implies $\log_{10} y = k$. (ii) Basic rules of logarithms e.g. $\log_{10}(pq) = \log_{10} p + \log_{10} q$ $\log_{10}(p/q) = \log_{10} p - \log_{10} q$ $\log_{10} p^n = n \log_{10} p$. (iii) Use of tables of logarithms and antilogarithms. 1.6 Sequence and Series (i) Patterns of sequences. (ii) Arithmetic progression (A.P.) Geometric Progression (G.P.) 1.7 Sets (i) Idea of sets, universal sets, finite and infinite sets, subsets, empty sets and disjoint sets. Idea of and notation for union, intersection and complement of sets. (ii) Solution of practical problems involving classification using Venn diagrams. 1.8 Logical Reasoning - Simple statements. True and false statements. Negation of statements, implications. 1.9 Positive and negative integers, rational numbers - The four basic operations on rational numbers. 1.10 Surds (Radicals) - Simplification and rationalization of simple surds. 1.11 Matrices and Determinants (i) Identification of order, notation and types of matrices. (ii) Addition, subtraction, scalar multiplication and multiplication of matrices. (iii) Determinant of a matrix 1.12 Ratio, Proportions and Rates - Ratio between two similar quantities. - Proportion between two or more similar quantities. - Financial partnerships, rates of work, costs, taxes, foreign exchange, density (e.g. population), mass, distance, time and speed.
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2	ALGEBRAIC PROCESSES	2.1 Algebraic expressions (i) Formulating algebraic expressions from given situations (ii) Evaluation of algebraic expressions 2.2 Simple operations on algebraic expressions (i) Expansion (ii) Factorization (iii) Binary Operations 2.3 Solution of Linear Equations (i) Linear equations in one variable (ii) Simultaneous linear equations in two variables. 2.4 Change of Subject of a Formula/Relation (i) Change of subject of a formula/relation (ii) Substitution 2.5 Quadratic Equations (i) Solution of quadratic equations (ii) Forming quadratic equation with given roots. (iii)Application of solution of quadratic equation in practical problems. 2.6 Graphs of Linear and Quadratic functions (i) Interpretation of graphs, coordinate of points, table of values, drawing quadratic graphs and obtaining roots from graphs. (ii) Graphical solution of a pair of equations of the form: $y = ax^2 + bx + c$ and $y = mx + k$ (iii) Drawing tangents to curves to determine the gradient at a given point. 2.7 Linear Inequalities (i) Solution of linear inequalities in one variable and representation on the number line. (ii) Graphical solution of linear inequalities in two variables. (iii) Graphical solution of simultaneous linear inequalities in two variables 2.8 Algebraic Fractions Operations on algebraic fractions with: (i) Monomial denominators ii) Binomial denominators 2.9 Functions and Relations -Types of Functions.
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3	MENSURATION	3.1 Lengths and Perimeters (i) Use of Pythagoras theorem, sine and cosine rules to determine lengths and distances. (ii) Lengths of arcs of circles, perimeters of sectors and segments. (iii) Longitudes and Latitudes. 3.2 Areas (i) Triangles and special quadrilaterals – rectangles, parallelograms and trapeziums. (ii) Circles, sectors and segments of circles. (iii) Surface areas of cubes, cuboids, cylinder, pyramids, right triangular prisms, cones and spheres. 3.3 Volumes (i) Volumes of cubes, cuboids, cylinders, cones, right pyramids and spheres. (ii) Volumes of similar solids
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4	PLANE GEOMETRY	4.1 Angles (i)Angles at a point add up to 360°. (ii)Adjacent angles on a straight line are supplementary. (iii) Vertically opposite angles are equal. 4.2 Angles and intercepts on parallel lines (i) Alternate angles are equal. (ii)Corresponding angles are equal. (iii)Interior opposite angles are supplementary (iv) Intercept theorem. 4.3 Triangles and Polygons (i) The sum of the angles of a triangle is 2 right angles. (ii) The exterior angle of a triangle equals the sum of the two interior opposite angles. (iii) Congruent triangles. (iv) Properties of special triangles - Isosceles, equilateral, right-angled, etc (v) Properties of special quadrilaterals – parallelogram, rhombus, square, rectangle, trapezium. (vi)Properties of similar triangles. (vii) The sum of the angles of a polygon (viii) Property of exterior angles of a polygon. (ix) Parallelograms on the same base and between the same parallels are equal in area. 4.4 Circles (i)Chords. (ii)The angle which an arc of a circle subtends at the centre of the circle is twice that which it subtends at any point on the remaining part of the circumference. (iii)Any angle subtended at the circumference by a diameter is a right angle. (iv)Angles in the same segment are equal. (v)Angles in opposite segments are supplementary. (vi)Perpendicularity of tangent and radius. (vii)If a tangent is drawn to a circle and from the point of contact a chord is drawn, each angle which this chord makes with the tangent is equal to the angle in the alternate segment. 4.5 Construction (i) Bisectors of angles and line segments (ii) Line parallel or perpendicular to a given line. (iii)Angles e.g. 90°, 60°, 45°, 30°, and an angle equal to a given angle. (iv) Triangles and quadrilaterals from sufficient data. 4.6 Loci Knowledge of the loci listed below and their intersections in 2 dimensions. (i)Points at a given distance from a given point. (ii)Points equidistant from two given points. (iii)Points equidistant from two given straight lines. (iv)Points at a given distance from a given straight line.
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5	COORDINATE GEOMETRY OF STRAIGHT LINES	(i) Concept of the x-y plane. (ii) Coordinates of points on the x-y plane.
6	TRIGONOMETRY	6.1 Sine, Cosine and Tangent of an angle (i) Sine, Cosine and Tangent of acute angles. (ii) Use of tables of trigonometric ratios. (iii) Trigonometric ratios of 30° , 45° and 60° . (iv) Sine, cosine and tangent of angles from 0° to 360° . (v) Graphs of sine and cosine. (vi) Graphs of trigonometric ratios. . 6.2 Angles of elevation and depression (i) Calculating angles of elevation and depression. (ii) Application to heights and distances. 6.3 Bearings (i) Bearing of one point from another. (ii) Calculation of distances and angles
7	INTRODUCTORY CALCULUS	(i) Differentiation of algebraic functions. (ii) Integration of simple Algebraic functions.

8	STATISTICS AND PROBABILITY	8.1 Statistics (i) Frequency distribution (ii) Pie charts, bar charts, histograms and frequency polygons (iii) Mean, median and mode for both discrete and grouped data. (iv) Cumulative frequency curve (Ogive). (v) Measures of Dispersion: range, semi inter-quartile/interquartile range, variance, mean deviation and standard deviation. 8.2 Probability (i) Experimental and theoretical probability. (ii) Addition of probabilities for mutually exclusive and independent events. iii) Multiplication of probabilities for independent events.
9	VECTORS AND TRANSFORMATION	9.1 Vectors in a Plane - Vectors as a directed line segment. - Cartesian components of a vector - Magnitude of a vector, equal vectors, addition and subtraction of vectors, zero vector, parallel vectors, multiplication of a vector by scalar. 9.2 Transformation in the Cartesian Plane Reflection of points and shapes in the Cartesian Plane. - Rotation of points and shapes in the Cartesian Plane. - Translation of points and shapes in the Cartesian Plane. - Enlargement

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Chapter Three

Recommended Textbooks

Adelodun A. A (2000) Distinction in Mathematics: Comprehensive Revision Text, (3rd Edition) Ado –Ekiti: FNPL.

Anyebe, J. A. B (1998) Basic Mathematics for Senior Secondary Schools and Remedial Students in Higher/ institutions, Lagos: Kenny Moore.

Channon, J. B. Smith, A. M (2001) New General Mathematics for West Africa SSS 1 to 3, Lagos: Longman.

David –Osuagwu, M. et al. (2000) New School Mathematics for Senior Secondary Schools, Onitsha: Africana – FIRST Publishers.

Egbe. E et al. (2000) Further Mathematics, Onitsha: Africana – FIRST Publishers.

Ibude, S. O. et al. (2003) Algebra and Calculus for Schools and Colleges: LINCEL Publishers.

Tuttuh – Adegun M. R. et al. (1997), Further Mathematics Project Books 1 to 3, Ibadan: NPS Educational

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Chapter Four NECO Grading System

The performance and grading of the SSCE candidates in the NECO examination can be determined by the grading system. The basic criteria for admission into a tertiary institution and employment in most organizations in Nigeria are 5 credits in Mathematics, English, and 3 other subjects related to the field of study, in a minimum of two sittings.

MARKS	GRADES	REMARKS
75-100	A1	Excellent
70-74	B2	Very Good
65-69	B3	Good
60-64	C4	Credit
55-59	C5	Credit
50-54	C6	Credit
45-49	D7	Pass
40-44	E8	Pass
39 and below	F9	Fail

It is very important to score at least 50% which will give you a C6 in the 4 compulsory subjects as well as your core subjects or you could have problems getting admission and may have to re-write the exams.

For example, if you want to study Medicine and Surgery, you should try hard to score high in Mathematics, English, Civic Education, and the trade subject you are writing and your core subjects which are: Physics, Biology, and Chemistry

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Hausa	Agricultural Science	Commerce
Yoruba	Health Education	Further Math
Arabic	Physics	Visual Arts
Fin. Accounting	Literature	Music
Geography	Economics	Food and Nutrition
French	I.R.S	

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